

Sterilization of *Amyelois transitella* (Lepidoptera: Pyralidae) by electron beam technology

Running Title:

eBeam Sterilization of Navel Orangeworm

Lars Laber¹, Charles Burks², Houston Wilson¹

¹ Department of Entomology, University of California – Riverside, 900 University Ave., Riverside,
CA 92521

² USDA, Agricultural Research Service, San Joaquin Valley Agricultural Sciences Center, 9611
South Riverbend Avenue, Parlier, CA 93648

**This pre-submission manuscript draft is not intended as the final version. It is included
with an R repository for this study to aid in describing the context of the data and the
way in which the data were collected.**

Abstract

Background

Use of the sterile insect technique (SIT) for control of arthropods relies on the mass release of
sterile individuals to suppress wild populations of the target pest. Insects are commonly sterilized
with gamma irradiation from either cobalt or cesium source material, which poses some human
health and environmental risks. One potentially safer alternative could be the use electron beam
radiation. In this study, we evaluated whether a commercial electron beam system could serve
as a practical non-nuclear alternative for sterilizing *Amyelois transitella* (Lepidoptera: Pyralidae),
a major pest of tree nuts in California. To do this, we tested the effects of different doses of

electron radiation (150-550 Gy) on longevity, flight ability, fecundity and egg viability of *A. transitella*.

Results

We were able to completely sterilize female *A. transitella* at 177 Gy and males at 310 Gy without any significant impacts on their longevity or flight ability, demonstrating for the first time that electron radiation can reliably sterilize this pest.

Conclusion

Our research shows that electron radiation can be effective in sterilizing insects without causing significant harm to their health and fitness, and in this way may provide a viable non-nuclear alternative with the potential for high throughput sterilization in an SIT program.

Keywords

Sterile insect technique, Activity, longevity, mating, fertility, fecundity

1 Introduction

In arthropod pest management, the sterile insect technique (SIT) relies on the introduction of large numbers of mass-reared and sterilized individuals to suppress wild populations of a target pest, and has been used worldwide for pest management over the past 100 years (Dyck, Hendrichs, and Robinson 2021). For instance, SIT has been used successfully against agricultural pests such as the pink bollworm (Lepidoptera: Gelechiidae: *Pectinophora gossypiella*), livestock pests such as the new world screwworm (Diptera: Calliphoridae: *Cochliomyia hominivorax*), and disease vectors such as the yellow fever mosquito (Diptera: Culicidae: *Aedes aegypti*) (Baumhover et al. 1955; Gato et al. 2021; Henneberry 2019; Knipling 1955). To date, most SIT

programs have relied upon the use of ionizing radiation, most commonly gamma radiation, to sterilize insects (Darden, Maeyens, and Bushland 1954; Fifield et al. 2021).

Ionizing radiation for SIT programs is usually produced by gamma irradiators using a cobalt-60 or cesium-137 source material (California Department of Food and Agriculture n.d.; Fifield et al. 2021). These sources are long-lasting and reliable but also come with a wide range of human health and environmental safety risks, as well as high replacement costs (Fan and Niemira 2020). As such, there has been growing interest to find viable non-nuclear alternatives to gamma irradiation for sterilization of insects, such as electron beam technology (Wang et al. 2023; Yun, Seung-Hwan et al. 2014).

An electron beam is a stream of accelerated free electrons that are focused on a specific area. In this form, electrons can damage DNA. This is achieved either directly, when electrons hit a DNA molecule, causing it to break apart, or by creating reactive oxygen species when reacting with water molecules in the vicinity of DNA (Egerton 2015). Electron beams can be used to sterilize various materials, surfaces and substances, such as surgical equipment, medicine, produce and even water (Fan and Niemira 2020; Fifield et al. 2021). They have comparatively low operational costs and consume less energy than similarly powered x-ray irradiators are safe to operate and have a relatively short treatment time.

Navel orangeworm (*Amyelois transitella*, Lepidoptera: Pyralidae) is a major pest of almonds, pistachios, and walnuts in California (Ferguson and Haviland 2016; Michelbacher and Davis 1961; Pathak, Maskey, and Rijal 2021). This moth overwinters as larvae or pupae in remnant nuts that are left behind after harvest (Wade 1961). In the spring, overwintering *A. transitella* complete their development, emerge as adults and will typically complete 3-4 generations per year (Higbee and Siegel 2009). *Amyelois transitella* is considered a pest of tree nuts since they deposit their eggs directly onto the new crop nuts, and the larvae that emerge feed on developing nut kernels, which can cause significant reductions in crop yield and quality (Wilson et al. 2020). Additionally, nuts infested by *A. transitella* are frequently associated with increased levels of *Aspergillus flavus*, a

fungus of major regulatory concern because of the carcinogenic aflatoxins that it produces (Palumbo et al. 2014; Robens and Cardwell 2003). As such, tree nut producers typically aim for <2% infestation rates, which requires intensive monitoring and management. Current management strategies for *A. transitella* include orchard sanitation (i.e. removal and destruction of remnant nuts), mating disruption, well-timed insecticide applications, and timely harvest²⁰. Recently, there has been growing interest in the development of an SIT program for *A. transitella* to provide an additional control strategy (California Department of Food and Agriculture n.d.; Higbee and Siegel 2009).

Here, we evaluated whether electron beam irradiation can reliably sterilize *A. transitella* without compromising traits important for a successful SIT program, such as insect longevity, flight ability and mating performance. While previous studies using electron radiation have focused primarily on insect mortality rather than sterilization (Alves et al. 2009; Hayashi et al. 2003; Jialin et al. 2011), we are here exploring the use of this technology as a non-nuclear alternative within an SIT program .

2 Methods

2.1 Navel Orangeworm (*Amyelois transitella*) Colonies

All *Amyelois transitella* (Lepidoptera: Pyralidae) moths were reared at the USDA Agricultural Research Service (ARS) San Joaquin Valley Agricultural Sciences Center (SJVASC) in Parlier, CA. Moths were reared at 26°C under a 14:10 L:D photoperiod with 60% relative humidity. Larvae were fed a standard wheat bran diet throughout their development (Tebbetts, Curtis, and Fries

1978). Unmated adult moths, separated by sex, were isolated within 36 h of eclosion for use in the irradiation assays.

2.2 Preparation of Moths for Irradiation and Dosimetry

Prior to irradiation, individual moths were arranged within a 127 × 82.55 mm 48-well plates (6 × 8 wells) (Corning®Costar®HTS Transwell®, CLS 4395, Corning, NY). To ensure even spread, optimal dose distribution and prolonged cooling, moths were placed individually into well-plate cells, with 5 moths per row, leaving one empty well in each row. To measure the applied radiation dose, an alanine tablet (FWT-50-30, DE582, Harwell/Steris, Didcot, Oxfordshire, UK) was placed into the empty well in every other row. We included two controls in the experiment. One control group was kept under colony conditions at the USDA-ARS San Joaquin Valley Agricultural Sciences Center, while the other control group was placed into the 48-well plates as described, and traveled with the other moths to the eBeam facility and back.

2.3 Transportation of Moths to the Commercial eBeam Facility

Following placement into the well plates, moths were cooled to 11.0 ± 1.0 °C to reduce movement and stress. The plates were then wrapped in multiple layers of cushioned foil (Bubblewrap®/Sealed Air Corporation, Charlotte, NC) and placed in an electric cooler (30l portable refrigerator, Euhomy, Sacramento, CA) set to 8 °C that included a temperature sensor (HOBO Temperature/RH Data Logger, MX2301A, Onset Computer Corporation, Bourne, MA) to confirm temperature during transport. After packaging, the moths were driven from the Kearney Agricultural Research and Extension Center (Parlier, CA) to the commercial eBeam facility (Steri-Tek, Fremont, CA), which was about a 3-hour drive away. Moths were then held in the cooler overnight until irradiation the following morning. Such trips were made on two separate occasions

for the fitness tests, and another two such occasions for the fertility tests (Table 1). These replicates in time are referred to as “Runs” in this paper.

2.4 Irradiation Procedures

At the eBeam facility, each well-plate was randomly assigned to one of the intended dose treatments (0, 200, 300, 400 and 500 Gy), then placed into Styrofoam racks ($26.67 \times 19.05 \times 2.54$ cm) that were then inserted into cardboard sleeves ($31.75 \times 24.13 \times 3$ cm) so they could be secured within the conveyor frames that moved through the electron beam. All sleeves were processed identically, and the control sleeve underwent the same handling procedure except for irradiation. During irradiation, the frames passed through the electron beam on a conveyor belt. The beam delivered a constant dose rate of 166 Gy/min, and for each treatment the applied dose was adjusted by changing the conveyor speed.

Following irradiation, all well-plates were removed, repackaged as before, and returned to the portable cooler. The full irradiation procedure in the commercial facility took approximately 4 hours, followed by 3 hours of driving back to the Kearney Agricultural Research and Extension Center. Upon arrival at the laboratory, moths were assigned to either (i) flight ability and longevity assays or (ii) mating and sterility assays in alternating experimental runs because the number of *A. transitella* available per run (300–360 individuals) was insufficient to conduct all assays simultaneously. Two runs were completed for each assay, with a total of 120 individuals (60 females and 60 males) evaluated per treatment and assay. Each run consisted of 30 individuals of each sex per treatment.

For flight ability and longevity assays, groups of 10 same-sex *A. transitella* were placed in a flight cage. Each flight cage constituted one replicate, resulting in three replicates (30 individuals) per treatment and sex per run.

For mating and sterility assays, 30 treated individuals of each sex and treatment were individually paired with untreated partners in each run. Each mating pair constituted a single replicate, resulting in 30 replicates per treatment and sex per run.

2.5 eBeam Dosimetry

As mentioned, dosimetry of the eBeam system was determined using alanine pellets irradiated alongside the moths in the same well-plates. The pellets were purchased from and read by the National Center for Electron Beam Research at Texas A&M University (College Station, TX) to determine absorbed dose. Tablets were shipped to Texas A&M University one day after irradiation. Alanine tablets were included in all well-plates brought to the eBeam facility, including the unirradiated control (i.e. Trip-Control treatment), but not in the control well-plates that remained in the laboratory (i.e. Lab-Control treatment).

2.6 Fitness Assay (Flight Ability and Longevity)

Adults from two trips to the eBeam facility were tested for flight and longevity (Table 1). Flight was evaluated using flight cylinders, a well-established assay for insect quality (Carpenter, Blomefield, and Vreysen 2012). Flight cylinders consisted of polyvinyl chloride (PVC) pipe (diameter = 16 cm, height = 16 cm) with interior surfaces coated with fluon (Insect-a-Slip™, Bioquip, Rancho Dominguez, CA) to ensure that moths escaped only by flight. Individual cylinders were placed within insect cages (30 × 30 × 30 cm; BugDorm-4E3030 Insect Rearing Cage, Bugdorm, Taiwan) and held under a summer photoperiod (14:10 L:D) at ambient temperature (26 ± 3°C, 60% RH). Moths were placed in cylinders in groups of 10, separated by sex and treatment. The number of

moths inside and outside the flight cylinders was recorded after 24 h, after which observations continued for the longevity assay.

For the longevity assay, moths were kept under the same conditions as in the flight assay (summer photoperiod 14:10 L:D; ambient temperature $26 \pm 3^{\circ}\text{C}$, 60% RH). The flight cylinders were removed, and water-soaked cotton balls (Jumbo Cotton Balls, Equate, Brandonville, AR) were added to all cages. These cotton balls were rehydrated every two days for the duration of the experiment. The number of surviving moths was noted daily. The experiment was terminated after 14 days.

2.7 Fertility Assay (Mating, Fecundity, and Larval Development)

Adults from the other two trips to the eBeam facility (Table 1) were used tested for reproductive ability. Individual moths were caged with a fertile, unmated moth of the opposite sex in plastic 30 mL portion cups and allowed to mate over a 24-hour period. The female individual of each pairing was then transferred to a 473 ml glass jar for the oviposition assay. The jars were covered with a round filter paper held in place with a banded lid and then placed on their sides so that the filter paper served as a vertical oviposition substrate.

Previous work has shown that after four days over 75% of eggs have been deposited (Burks 2014). In addition, *A. transitella* is known to cannibalize other larvae (Bush et al. 2017), so if the first larvae emerge and no diet is present, they could start feeding on each other, changing the results obtained. Therefore, after three days or at time of death female moths were retained individually pending examination for mating status and the oviposition paper was removed and the eggs counted. The paper was then placed egg-side down on wheat bran diet in a petri dish. The *A. transitella* eggs were allowed to incubate for 25 days at 26°C , 60 % RH and a 14:10 h (L:D) photoperiod.

To determine mating status, the bursa copulatrix of all female moths was dissected to confirm that a single spermatophore was present. To determine fecundity, all eggs deposited onto the filter paper were recorded. Finally, to determine larval development success, the number of pupae per petri dish was noted after the 25-day incubation period.

2.8 Data Analysis

All data was analyzed using the R Statistical Program, version 4.5.2 (R Core Team 2024). Other packages used glmmTMB (McGillucuddy et al. 2025) to fit and compare generalized linear models, DHARMa (Hartig et al. 2026) to examine model fit, car (Fox and Weisburg 2019) to conduct type II analysis of deviance using likelihood ratio or Wald chi-square tests. Generalized linear models (GLMs) were used to analyze the data, response variables including the proportion of moths escaping from the flight cylinder, the proportion of moths that survived > 14 days, the proportion of moths mated, the number of eggs laid, and the number of F1 that developed to pupa were analyzed using a generalized linear model (GLM) with dose, sex, run, and all two-way interactions as predictors. For the flight cylinder and longevity analyses, the cage containing 10 adults was the experimental unit. A quasibinomial distribution was used with the GLM, and dose (Gy) was modeled as a continuous variable. In addition, the data were plotted with linear regression lines including the 95% confidence interval of the mean in order to illustrate trends.

For analyses of fertility assays for the other two runs (with mating, fecundity, and pupal count as response variables) the experimental unit was individual adults. The proportion of adults that mated was modeled with dose as a continuous variable, using a GLM with a binomial distribution. However, for egg and pupa counts, doses were analyzed as a categorical variable in order to better fit models to data with overdispersed frequency and complete separation. Doses of 176 and 177 Gy were both categorized and described as “175 Gy”, doses of 208 and 211 Gy were

categorized and described as “210 Gy”, and doses of 405 and 445 Gy were categorized and described as “420 Gy”.

For analysis of the oviposition and larval development assays, we only used data from mated individuals and rejected all data gathered from individuals who were found to have not mated. Oviposition data were analyzed using a GLM with a negative binomial distribution, and counts of F1 that developed to the pupal stage were compared using a GLM with a zero-inflated negative binomial distribution. Based on findings from the GLMs, counts were compared between doses separately within each level of sex $\times$ run. The non-parametric Kruskal-Wallis test was used to determine if there were significant differences between levels of dose, and the Dunn test with the Holm adjustment for multiple comparisons was used to compare between doses.

3 Results

3.1 Irradiation and Dosimetry

As mentioned, conveyor belt speed was adjusted to achieve the desired irradiation treatments of 100, 200, 300, 400 and 500 Gy. Readings from the alanine tablets indicated that the applied doses were close to, but not exactly, at the levels desired. A comparison of the intended and applied treatment doses are provided in Table 1. Regardless, a low-to-high range of electron beam radiation doses were successfully applied to the moths.

3.2 Flight Ability and Longevity Assays

3.2.1 Flight Ability Assay

Flight ability was not affected by increasing electron beam dose (Table 2), in either female or male *A. transitella* (Fig. 1). Across all doses, males escaped from flight cylinders more often than females (males 7.19 ± 0.26 out of 10, females 3.97 ± 0.37 out of 10; $t(62.8) = 7.031$, $P < 0.001$).

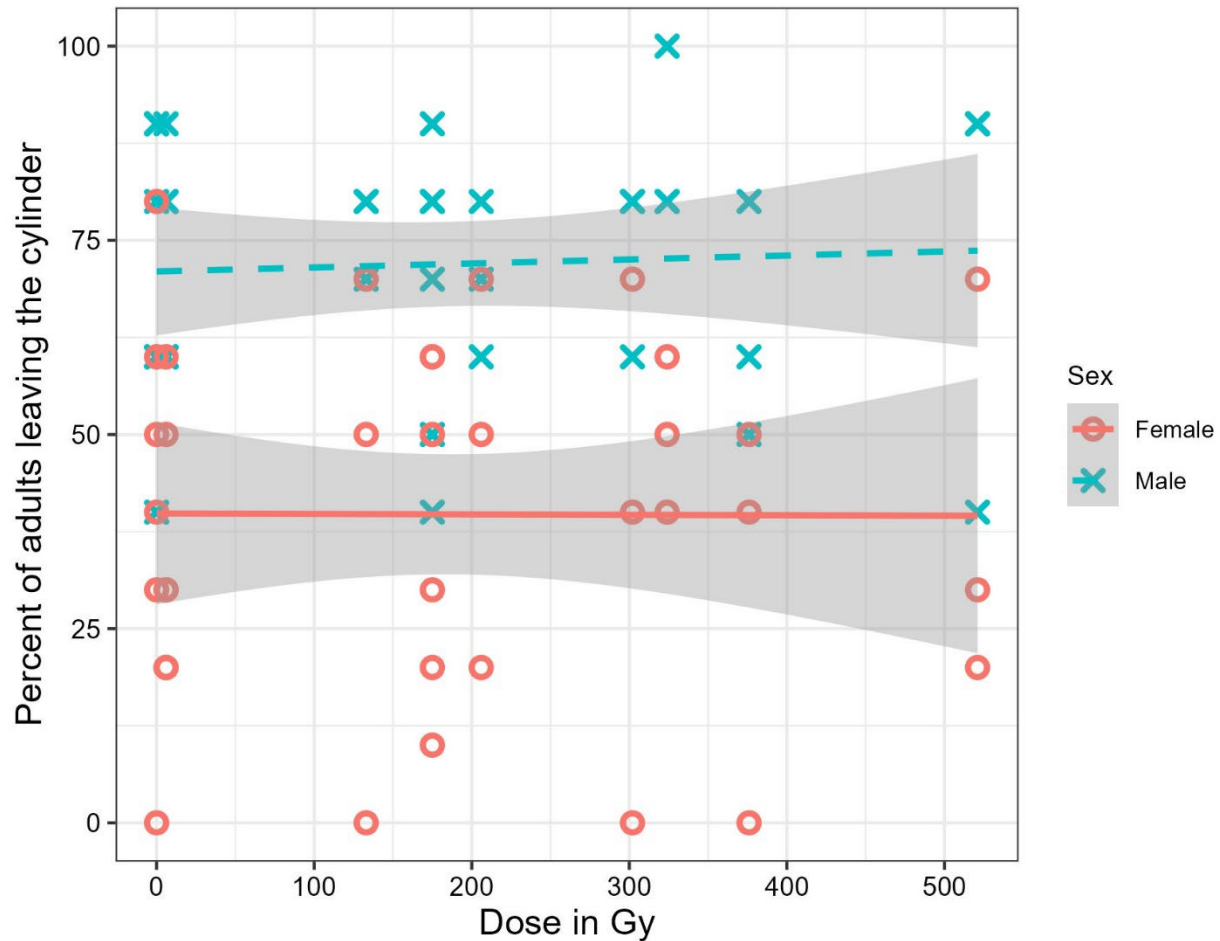


Figure 1 Effect of eBeam irradiation indicated by dose (Gy) on flight ability of female and male *A. transitella*. Symbols indicate individual flight cylinder experiments (groups of 10 moths). The lines indicate the linear regression fit for each sex, and the grey area indicated the 95% prediction interval.

3.2.2 Survivorship Assay

Dose did not have a significant effect on longevity (Table 3). There was a significant effect of dose on longevity, the dose * sex interaction was significant, and the effect of

run (cohort) was marginally significant (Table 3, Fig. 2). More females than males lived longer than 14 d ($33 \pm 3.2\%$ vs $23 \pm 3.1\%$).

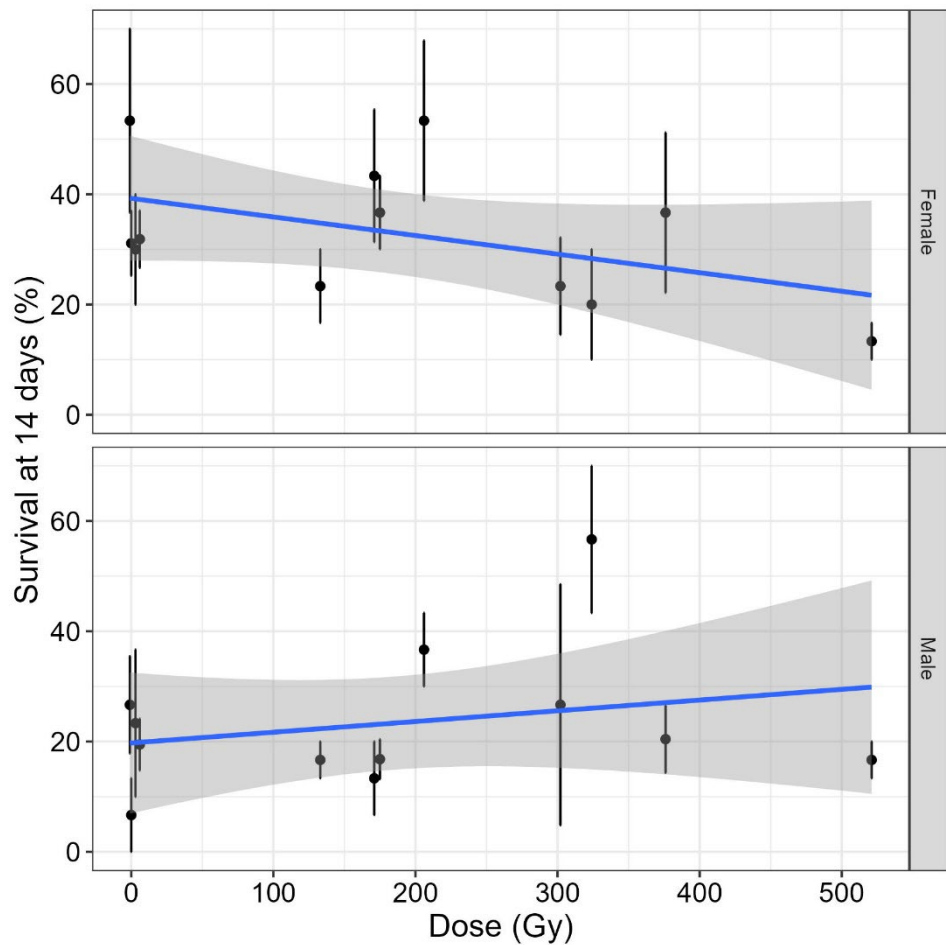


Figure 2 Effect of eBeam irradiation indicated by dose (Gy) on the proportion of *A. transitella* that survived > 14 d during the first (green x's) and second (purple rings) runs of the experiment. Green crosses and purple squares symbolize the lab controls. The lines indicate the linear regression fit for each sex, and the grey area indicated the 95% prediction interval.

3.3 Mating, Oviposition, and Development Assay

3.3.1 Mating Assay

The effect of sex on the proportion of moths that mated was marginally significant (Table 4). Other effects, including dose, were not significant (Table 4, Figure 3).

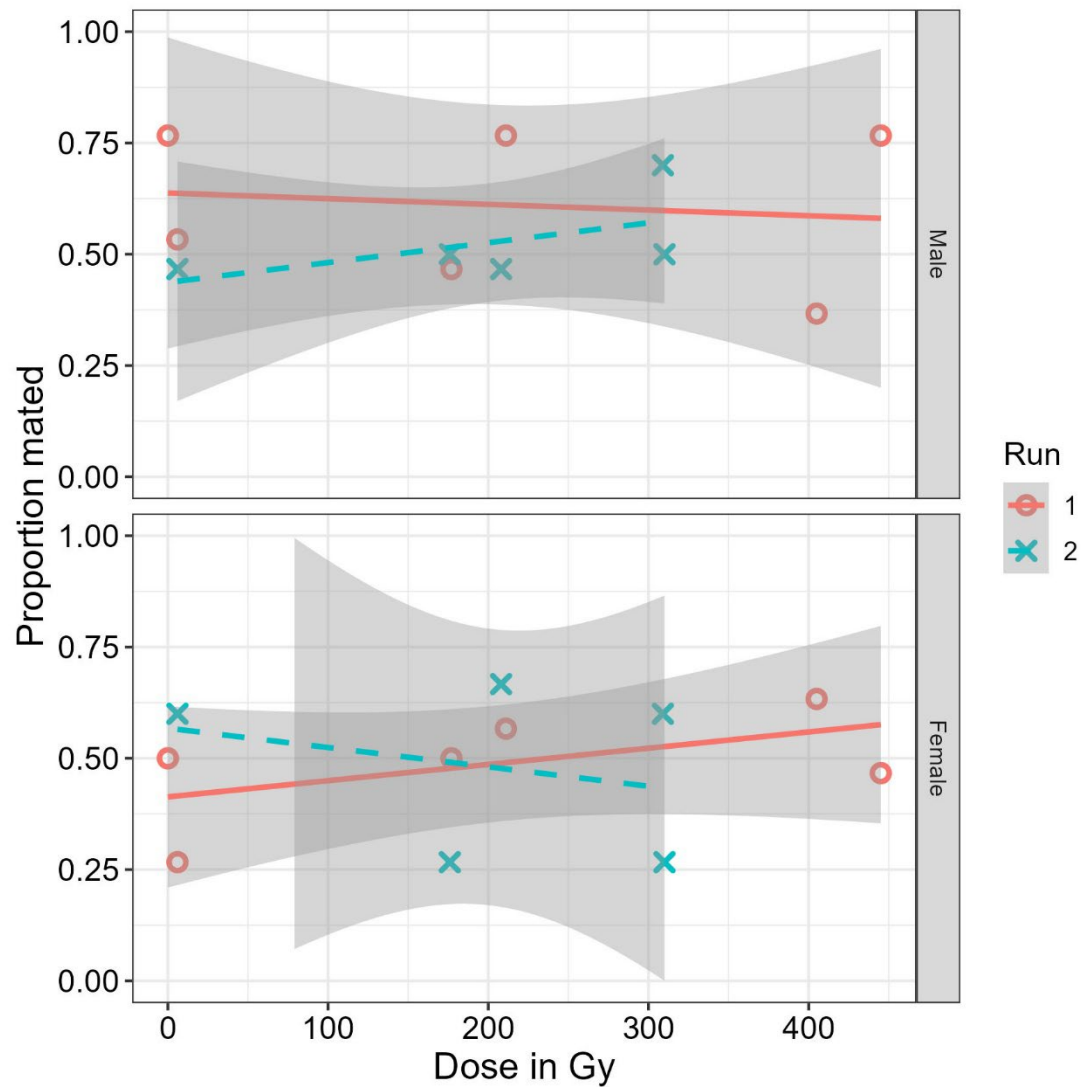


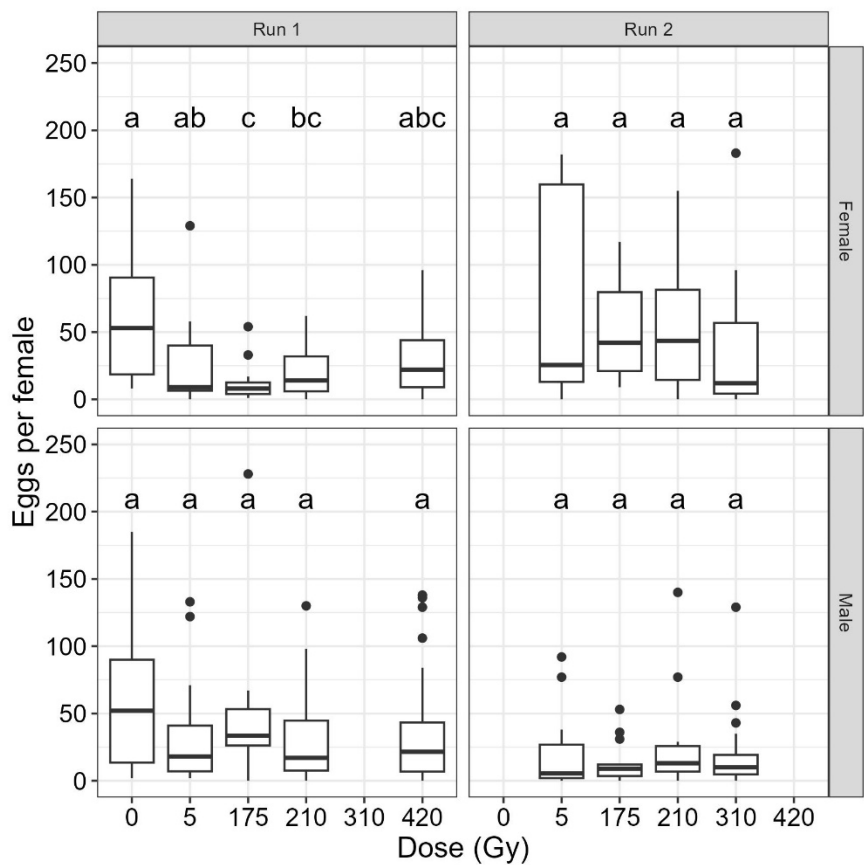
Figure 3 Effect of eBeam irradiation indicated by dose (Gy) on the mating ability of female and male *A. transitella* indicated by the proportion of mated individuals after two days in the first (red rings) and second (cyan x's) runs with an untreated individual of the opposite sex out of groups of ten. The linear regression lines and the 95% prediction intervals for each level of sex * run are respectively indicated by the straight lines and shaded areas.

266

267 3.3.2 Oviposition Assay

268 There were significant effects on total oviposition per female due to dose, and there was
269 also a significant interaction of sex * run (Table 5). Within sex * sun, differences between
270 doses tested were only significant for females in Run 1 (Fig. 4). Eggs were produced from
271 pairings for over 90% of mated adults tested, and within sex * run there was no significant
272 difference in eggs per female between the lowest and highest doses.

273



274

275 **Figure 4** Box plots illustrating the effect of eBeam irradiation on the fecundity of *A. transitella*. Sex indicates
276 the sex irradiated; for “Male” this means eggs per non-irradiated female paired with an irradiated male.
277 Treatment groups in the same panel with different letters are significantly different. The 0 treatment
278 indicates moths that were not taken to the eBeam facility, whereas the 5 Gy treatment indicates moths that
279 were taken to the eBeam facility but not exposed.

3.3.3 Development Assay

The GLM examining effects on the count of pupae produced by experimental pairings revealed highly significant effects due to dose and the sex * run interaction, in addition to significant effects of run and dose*run and a marginally significant effect due to dose*sex. When treatment differences were compared within sex * run, dose dependent effects were evident (Fig. 5). This dose-dependence was generally more incremental when displayed as percent fertile (i.e., percent of pairings that produced pupae) (Fig. 5). The lowest doses resulting in no pupae were 210 Gy for females (0 out of 37 pairs), and 310 Gy for males (0 out of 36 pairs).

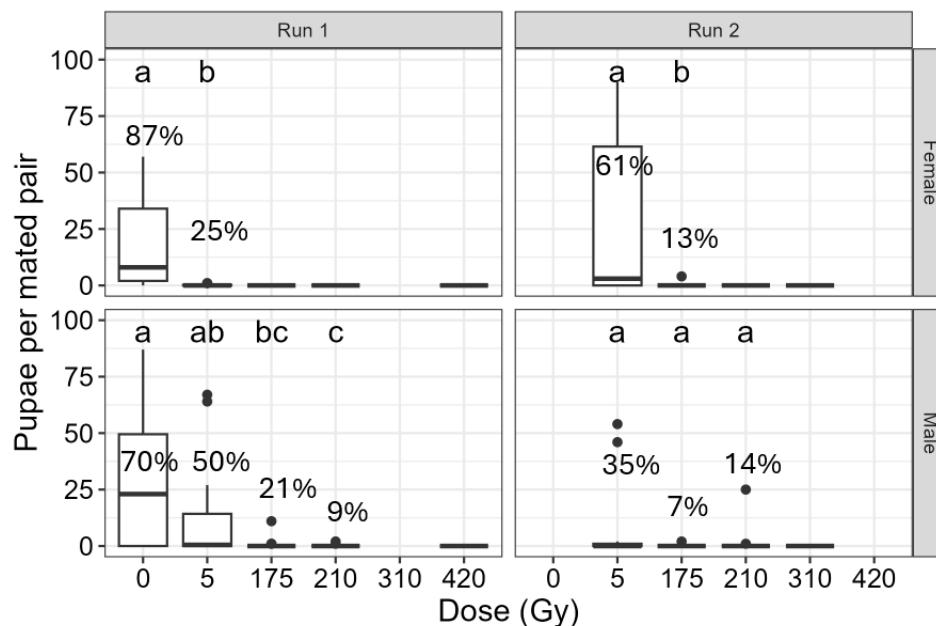


Figure 5 Box plots indicating pupae per female following successful mating. Treatment groups in the same panel with different letters are significantly different (treatments with 0 pupae were not included in this analysis). The numerals indicate the percent of pairings that produced > 0 pupae.

296 4 Discussion

297 Our findings demonstrate that electron beam irradiation of *A. transitella* does not
298 adversely affect flight ability, longevity, mating success, or fecundity. Although some
299 significant differences among individual dose treatments were observed, no dose-
300 dependent trends were evident for these parameters. In contrast, fertility declined
301 consistently with increasing electron beam dose. Complete sterility was achieved at
302 doses >200 Gy for females and >300 Gy for males, without compromising other fitness
303 traits. The greater radio tolerance of males compared to females has been documented
304 across numerous insect species (Stephanie Bloem, Carpenter, and Bloem 2003; Boshra
305 and Mikhael 2006; Flint and Merkle 1983). Since no detrimental effects beyond reduced
306 fertility were observed with females who received >300 Gy, it may be possible to sterilize
307 both males and females simultaneously using the higher dose required for male sterility.
308 Doses below the sterilizing threshold still reduced fertility substantially, although a small
309 number of offspring did successfully develop. Similar results have been reported for X-
310 ray irradiation of *A. transitella* before, where fertility reductions exceeded 99%, yet
311 complete sterility required considerably higher doses (Light et al. 2015). Individual
312 variation in radiotolerance or DNA repair capacity may explain the survival of these rare
313 offspring (Bakri et al. 2005; LaChance and Graham 1984; Yamada et al. 2022).
314 Alternatively, these observations may reflect stochastic variation in radiation-induced
315 damage (Yamada et al. 2022). Given the well-documented phenomenon of inherited
316 sterility in Lepidoptera, it is possible that surviving offspring would themselves exhibit
317 reduced fertility, or even be fully infertile (S. Bloem et al. 1999; Carpenter, Bloem, and

Marec 2005; Marec, Kollárová, and Pavelka 1999). Unfortunately, F1 sterility was not able to be evaluated here.

Irradiated *A. transitella* remained viable and capable of flight for more than seven days at sterilizing doses exceeding 500 Gy, which provides sufficient time for dispersal and mating (Parra-Pedrazzoli and Leal 2006). Median survival was greater than that reported in previous X-ray studies (Light et al. 2015), suggesting that electron beam irradiation may cause less tissue damage. This difference may result from the higher energy levels (higher kV) used in electron beam systems, which can improve penetration depth and increase the proportion of energy directed toward DNA damage rather than localized tissue damage (Ponomarev 2024; Russ 1915). Higher beam energies may therefore improve both insect quality and processing efficiency, which are important considerations for scaling up these processes in an areawide SIT program.

While electron beam irradiation proved highly effective for sterilizing *A. transitella*, we also found that handling, transportation, and associated stress appeared to reduce fertility in untreated controls brought to the commercial facility, compared with controls that remained in the laboratory. While different methods of rearing and handling *A. transitella* can affect their performance (Reger et al. 2020, 2021), this is a separate issue from determination of the sterilizing dose, which was our goal here. Still, as with any SIT program, the rearing, handling, sterilization, transport and release procedures for *A. transitella* will all need to be optimized to develop an areawide program. Here, as a first step, we demonstrated the potential of electron beam irradiation as a high-throughput method of sterilization for *A. transitella* with minimal impacts on performance.

5 Conclusion

The results presented in this study show that it is possible to use a commercially available electron beam facility to successfully sterilize *A. transitella*. Electron beam technology can deliver a higher throughput without negatively affecting *A. transitella* compared to currently available X-ray irradiators. Based on these results, electron beam technology may be a viable alternative to X-ray sterilization, when considering options for the replacement of gamma irradiators in SIT programs.

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7 Conflict of Interest Declaration:

The authors declare no conflicts of interest

8 Author contribution statement

- Conceptualization – LL, CB, HW
- Data curation – LL
- Formal analysis – LL, CB
- Funding acquisition – HW, CB
- Investigation – LL, CB

- 367 • Methodology – LL, CB, HW
- 368 • Project administration – CB, HW
- 369 • Resources – CB, HW
- 370 • Software – HW
- 371 • Supervision – CB, HW
- 372 • Validation – LL, CB
- 373 • Visualization – LL
- 374 • Writing – original draft – LL, CB
- 375 • Writing – review & editing – LL, HW, CB
- 376

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10 Tables

Table 1 Mean measured dose($\pm$ SE) in Gy of six alanine tablets per intended dose, run and experiment and how the doses are labeled in the following figures to enhance visual clarity. Both lab controls and trip controls were not exposed to irradiation, therefore lab controls of first runs (*) are marked with green crosses instead of x's, lab controls of second runs (**) are depicted as purple squares instead of purple rings. For lab controls, no alanine dose readings were conducted. For run 2 of the fertility experiment (***), due to lack of emerging moths, no lab control could be included in the experiment.

Intended Dose	Fitness				Fertility			
	Measured Dose (Gy)		Depicted Dose (Gy)		Measured Dose (Gy)		Depicted Dose (Gy)	
	Run 1	Run 2	Run 1	Run 2	Run 1	Run 2	Run 1	Run 2
Lab Control	-	-	0*	0**	0	-***	0*	-
Trip Control	0	0	0	0	0	0	0	0
200 Gy	134.0 $\pm$ 1.9	171.3 $\pm$ 1.1	134	171	177.3 $\pm$ 0.7	176.4 $\pm$ 0.5	177	176
300 Gy	175.0 $\pm$ 0.8	205.7 $\pm$ 1.1	175	206	210.7 $\pm$ 0.9	208.0 $\pm$ 0.6	211	208
400 Gy	302.0 $\pm$ 0.8	323.7 $\pm$ 5.8	302	324	405.0 $\pm$ 0.3	310.0 $\pm$ 1.2	405	311
500 Gy	376.0 $\pm$ 3.0	520.7 $\pm$ 3.6	376	521	445.5 $\pm$ 15.0	309.5 $\pm$ 1.7	446	310

Table 2 Analysis of deviance table, showing type II likelihood ratio chi-square tests of main effects and all two-way interactions for the proportion of adults in experiment 1 that escaped from the flight tunnels (GLM with quasibinomial distribution, residual df = 65).

Effect	χ^2	df	P
Dose (Gy)	0.001	1	0.98
Sex	43.83	1	< 0.001
Run (Cohort)	2.38	1	0.12
Dose $\times$ Sex	0.061	1	0.80
Dose $\times$ Run	0.129	1	0.72
Sex $\times$ Run	0.018	1	0.89

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Table 3 Analysis of deviance table, showing type II likelihood ratio chi-square tests of main effects and all two-way interactions for the proportion of adults in experiment 1 that survived >14 days (GLM with quasibinomial distribution, residual df = 65).

Effect	χ^2	df	P
Dose (Gy)	0.650	1	0.42
Sex	5.056	1	< 0.025
Run (Cohort)	3.761	1	0.052
Dose × Sex	3.870	1	0.049
Dose × Run	2.439	1	0.118
Sex × Run	0.674	1	0.412

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Table 4 Analysis of deviance table, showing type II likelihood ratio chi-square tests of main effects and all two-way interactions for the proportion of adults in experiment 2 that mated (GLM with binomial distribution, residual df = 653).

Effect	χ^2	df	P
Dose (Gy)	0.150	1	0.698
Sex	2.996	1	0.083
Run (Cohort)	0.576	1	0.448
Dose × Sex	0.271	1	0.602
Dose × Run	0.144	1	0.704
Sex × Run	0.988	1	0.320

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Table 5 Analysis of deviance table, showing type II likelihood ratio chi-square tests of main effects and all two-way interactions for the total number of eggs laid by treated adults in experiment 2 (GLM with negative binomial distribution, dose categorical, residual df = 333).

Effect	χ^2	df	P
Dose (Gy)	23.5	1	< 0.001
Sex	2.57	1	0.12
Run (Cohort)	0.92	1	0.34
Dose × Sex	5.73	5	0.33
Dose × Run	0.59	2	0.74
Sex × Run	21.3	1	< 0.001

11 Figure Legends

Figure 1 Effect of eBeam irradiation indicated by dose (Gy) on flight ability of female and male *A. transitella*. Symbols indicate individual flight cylinder experiments (groups of 10 moths). The lines indicate the linear regression fit for each sex, and the grey area indicated the 95% prediction interval.

Figure 2 Effect of eBeam irradiation indicated by dose (Gy) on the proportion of *A. transitella* that survived > 14 d during the first (green x's) and second (purple rings) runs of the experiment. Green crosses and purple squares symbolize the lab controls. The lines indicate the linear regression fit for each sex, and the grey area indicated the 95% prediction interval.

Figure 3 Effect of eBeam irradiation indicated by dose (Gy) on the mating ability of female and male *A. transitella* indicated by the proportion of mated individuals after two days in the first (red rings) and second (cyan x's) runs with an untreated individual of the opposite sex out of groups of ten. The linear regression lines and the 95% prediction intervals for each level of sex * run are respectively indicated by the straight lines and shaded areas.

Figure 4 Box plots illustrating the effect of eBeam irradiation on the fecundity of *A. transitella*. Sex indicates the sex irradiated; for “Male” this means eggs per non-irradiated female paired with an irradiated male. Treatment groups in the same panel with different

letters are significantly different. The 0 treatment indicates moths that were not taken to the eBeam facility, whereas the 5 Gy treatment indicates moths that were taken to the eBeam facility but not exposed.

Figure 5 Box plots indicating pupae per female following successful mating. Treatment groups in the same panel with different letters are significantly different (treatments with 0 pupae were not included in this analysis). The numerals indicate the percent of pairings that produced > 0 pupae.

12 Data availability statement

Datasets and analysis scripts used in this study are available from the authors on reasonable request.

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